

Research Ethics Timeline

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 - **Research Ethics Timeline**
 - What is Ethics in Research & Why is it Important?

David B. Resnik, J.D., Ph.D., Bioethicist,
NIEHS/NIH
Florian W. Hofweber

Note: This list is the authors' own interpretation of some important events in the history of research ethics and does not include every event that some people might regard as important. We welcome suggestions for additions, revisions, etc. Please send them to [David B. Resnik](#).

- **1500s**

Mughal emperor Akbar the Great performs an experiment to determine whether children who grow up in a mute environment will learn language. He ordered twelve infants to be raised by mute nurses who communicated with each other via sign language. He later came back to discover that the twelve children did not learn an audible language but instead communicated in sign. Similar experiments have been done by other monarchs, many with the purpose of discovering the “original” language.

- **1620**

Philosopher and scientist Francis Bacon (1561-1626) publishes *The Novum Organon*, in which he argues that scientific knowledge should be based on observation and experimentation and not on tradition and authority and that it should benefit humanity.

- **1632**

Physicist and astronomer Galileo Galilei (1564-1642) publishes his *Dialogue on Two World Systems*, in which he defends a heliocentric theory of the solar system, a view that contradicted the Catholic Church's position that the Earth does not move but that the Sun moves around it. In 1633, Galileo appeared before an inquisitor from the Catholic Church. He was ordered to recant his views and was sentenced to house arrest for the remainder of his life. The Church banned his book. In 1992, 359 years after Galileo's arrest, Pope John Paul II formally apologized for its treatment of Galileo.

- **1662**

The Royal Society of London for Improving Natural Knowledge, the world's first scientific organization, is established for the purpose of realizing Bacon's vision of science. The Royal Society publishes the world's first scientific journal *Philosophical Transactions of the Royal Society of London*, in 1665.

- **1752**

The Royal Society of London institutes peer review procedures for articles submitted to *Philosophical Transactions*.

- **1796**

English physician Edward Jenner (1749-1823), the "father" of immunology, tests a vaccine for smallpox on an 8-year-old boy, James Phipps. Jenner and others had observed that dairymaids did not get smallpox. Jenner theorized that this was because they were exposed to cowpox. Jenner tested this hypothesis by inoculating Phipps with cowpox and then exposing him to smallpox. Phipps did not get smallpox. During Jenner's time, 10-20% of the global population died from smallpox each year.

- **1830**

English mathematician and inventor Charles Babbage (1791-1871) publishes *Reflections on the Decline of Science in England, And Some of Its Causes*, in which he argues that many of his colleagues were engaging in dishonest research practices, including fabricating, cooking, trimming, and fudging data. Babbage invented a programmable, mechanical computing machine.

- **1856**

Charles Darwin (1809-1882) and Lord Alfred Wallace publish *The Origin of Species*, which proposes a theory of evolution of living things by natural selection. The book generates a great deal of controversy because it proposes that human beings were not created by God (as most religions claimed) but descended from apes. Darwin collected most of the data for the theory while serving as the ship's naturalist on the voyage of the HMS Beagle (1831-1836). He waited over twenty years to publish his ideas because he knew they would meet with strong opposition and he wanted to ensure that he could back up his claims with evidence and arguments. George Lyell urged Darwin to publish his theory after reading a paper by Alfred Wallace that proposed a theory similar to Darwin's, so that Darwin could establish priority. Instead, Darwin shared credit with Wallace.

- **1874**

Roberts Bartholow (1831-1904) was treating a mentally disabled patient, 30-year-old Mary Rafferty, who had a two-inch hole in her skull caused by a cancerous ulcer. He inserted electrodes into the hole to study the effects of electrical stimulation on her brain. Rafferty experienced pain, distress, convulsions, and seizures, and fell into a coma and died in a few days.

- **1876-1957**

The Massachusetts Supreme Court rules in *McDonald v. Massachusetts General Hospital* that a charitable hospital is not liable for the actions of its employees. This was an important case for human research ethics because courts that followed its reasoning shielded hospitals from legal liability from medical experiments performed by employees. Later, in *Schloendorff v. Society of New York Hospital* (1914), the New York Court of Appeals ruled that Mary Schloendorff could sue her doctors, but not the hospital, for unconsented surgery. This legal principle, known as the Schloendorff rule, stood until 1957, when the New York Court of Appeals overturned it in *Bing v. Thunig*.

- **1885**

French chemist and microbiologist Louis Pasteur (1822-1895) tested a rabies vaccine on a 9-year-old boy, Joseph Meister, who had been attacked by a rabid dog two days before. The vaccine worked, and by 1886 Pasteur was treating hundreds of people. The vaccine had been tested on dogs but not humans. Pasteur could have been prosecuted for vaccinating people because he was not a licensed physician.

- **1897**

Giuseppe Sanarelli (1864-1940) injects bacteria into five patients without their consent to test his hypothesis that the bacteria cause yellow fever. The patients all developed yellow fever symptoms and three died. However, the hypothesis later was proven false by Walter Reed. Many physicians sharply criticized Sanarelli's experiments as being immoral.

- **1900**

U.S. Army physician Walter Reed (1851-1902) conducts medical experiments in Cuba in the early 1900s which show that *Aedes aegypti* mosquitos carry yellow fever. Yellow fever had been a major public health problem in the Caribbean and Central America that killed thousands of people each year and threatened commerce and U.S. military operations. Reed asked participants who had never had yellow fever to allow themselves to be bitten by mosquitos that had fed on patients with yellow fever, or to be injected with blood from a yellow fever patient. The participants signed informed consent forms (believed to be the first know use of this documentation) stating that they understood the risks of the experiment, including the possibility of death. The forms were translated into Spanish. Participants received \$100 in gold and additional \$100 and free medical care if they contracted yellow fever. Family members of participants who died also received \$100. Two of Reed's collaborators, James Carroll and Jesse Lazear, volunteered for the experiment. They both developed yellow fever and Lazear died. Reed had been planning to volunteer for the experiment, but Carroll talked him out of it due to his age (the disease was more deadly for patients over 40 years old, such as Reed). A total of 33 volunteers participated in the experiment, including 18 Americans (2 civilians and 16 soldiers) and 15 Spanish immigrants. Six people died from yellow fever. The surviving military personnel received medals and government pensions, and the Army's Medical Center in Washington, DC was named after Reed.

- **1909**

Robert Millikan (1868-1953) performs oil drop experiments to determine the charge of an electron. Millikan received a Nobel Prize for this research in 1923. Historians and journalists who studied Millikan's notebooks discovered that he did not report 33 out of 149 oil drop observations that he had marked as "fair" or "poor." Millikan also did not name his student, Harvey Fletcher, as an author on the paper that reported the results of these experiments, even though Fletcher made important contributions to the design of these experiments, such as suggesting that Millikan use oil droplets instead of water droplets.

- **1912**

Museum curator Charles Dawson discovers a skull in at Piltdown gravel bed near Surrey, U.K. It was thought to be the fossilized remains of a species in between humans and apes (i.e., “a missing link”). A controversy surrounded the skull for decades and many scientists believed it to be fake. Chemical analyses performed in 1953 confirmed these suspicions by showing that the skull is a combination of a human skull and orangutan jaw, which had been treated with chemicals to make them appear old. The identity of the forger is still unknown, though most historians suspect Dawson.

- **1925**

The University of Wisconsin establishes the Wisconsin Alumni Foundation (WARF), an independent organization that manages intellectual property (e.g. patents) and investments owned by the university and supports scientific innovation and discovery on campus. At that time, few universities owned or managed patents. WARF helps Harry Steenbock develop his invention for fortifying fats with vitamin D.

- **1932-1972**

The Tuskegee Syphilis Study, sponsored by the U.S. Department of Health, Education and Welfare, begins in 1932. The study investigated the effects of untreated syphilis in 400 African American men from the Tuskegee, Alabama area. The researchers did not tell the subjects that they were in an experiment. Most subjects who attended the Tuskegee clinic thought they were getting treatment for “bad blood.”

Researchers withheld treatment for the disease from participants even when penicillin, an effective form of treatment, became widely available in the 1950s. The study ended in 1972, after a news story from the Associated Press alerted the public and Congress to the ethical problems with the research. The U.S. government settled a lawsuit brought by the participants and their families. In 1997, President William Clinton issued an official apology on behalf of the U.S. government to surviving participants and their families.

- **1932-1945**

Japanese scientists working at Unit 731 performed morally abominable experiments on thousands of Chinese prisoners of war, including biological and chemical weapons experiments, vaccination experiments, and wound-healing and surgical studies, including vivisections. The U.S. government agreed to not prosecute the scientists for war crimes in exchange for data from the biological and chemical weapons research. Unit 731 of the Imperial Japanese Army also conducted research on Korean prisoners/civilians (such as Dong Ju Yoon (arguably the most famous modern era Korean poet) and Chung-Chun Lee (a Korean national hero and freedom

fighter)), as well as Mongolians, Manchurians (separate from Chinese), and Russians.

- **1939-45**

German scientists conducted morally abominable research on concentration camp prisoners, including experiments that exposed subjects to freezing temperatures, low air pressures, ionizing radiation and electricity, and infectious diseases; as well as wound-healing and surgical studies. The Allies prosecuted the German scientists for war crimes in the Nuremberg Trials. The Nuremberg Code provided the legal basis for prosecuting the scientists.

- **1940**

Two German refugee scientists, Frisch and R.E. Peierls, warn the U.S. about Germany's nuclear weapons program. Physicist Albert Einstein (1879-1955) sends a letter to President Roosevelt warning him about the threat posed by Germany. The letter, which was written by Leó Szilárd in consultation with Edward Teller and Eugene Wigner, was signed by Einstein. The letter suggested that the U.S. should develop a nuclear weapons program.

- **1942-1945**

The U.S. conducts the \$2 billion (\$34 billion in 2023 dollars) Manhattan Project to develop an atomic bomb. General Leslie Groves directs the Project and physicist Robert Oppenheimer (1904-1967) oversees the scientific work. Other notable scientists who worked on the project included Hans Beth (1906-2005), Enrico Fermi (1901-1954), Richard Feynman (1918-1988), and Edward Teller (1908-2003). At the time, the negative health effects of radiation were poorly not well-understood and the scientists working on the project were exposed to extremely large doses of radiation through the handling of plutonium and uranium. The first atomic bomb was detonated in the Jornada del Muerto Desert in New Mexico on July 16, 1945.

- **1944-1980s**

The U.S. Department of Energy sponsors secret research on the effects of radiation on human beings. Subjects were not told that they were participating in the experiments. Experiments were conducted on cancer patients, pregnant women, and military personnel. These experiments included in total several hundred releases of radiation on human subjects. They were often done to test weaponry or safety equipment. These experiments were investigated decades after they happened by the Advisory Committee on Human Radiation Experiments, which was created in 1994 by President William Clinton after he declassified documents

pertaining to this research. This committee also discovered that at least several hundred Uranium miners died of lung cancer, partly as a result of the government failing to properly ventilate the mines.

- **1945**

The U.S. drops atomic bombs on Hiroshima and Nagasaki, Japan on August 6 and 9, killing an estimated 200,000 civilians. After the bombs were dropped, Oppenheimer and other scientists led the “atoms for peace” movement.

- **1945**

Engineer and Head of the U.S. Office of Scientific Research and Development Vannevar Bush (1890-1974) writes the report *Science: The Endless Frontier* for President Franklin Roosevelt. The report argues for a major increase in government spending on science and defends the ideal of a self-governing scientific community free from significant public oversight. It advocates for investment in science and technology as a means of promoting national security and economic development.

- **1946**

The U.S. Congress passes a law transferring atomic energy development from military to civilian control. The law leads to the formation of the Atomic Energy Commission, which promotes peaceful uses of nuclear energy.

- **1947**

The Nuremberg Code, the first international code of ethics for research on human subjects, is adopted. The Code requires that research cannot take place without the subject’s consent; that research must be scientifically well-designed and socially valuable; and that research must minimize harm and suffering and should not involve a significant risk of death or disabling injury.

- **1947**

Norbert Wiener, the founder of cybernetics, published an article in the *Atlantic Monthly* titled “A Scientist Rebels” in which he refuses to conduct research for the military.

- **1948**

Alfred Kinsey (1894-1956) publishes *Sexual Behavior in the Human Male*. Five years later, he publishes *Sexual Behavior in the Human Female*. These books were very controversial, because they examined topics which were regarded as taboo at the time, such as masturbation, orgasm, intercourse, promiscuity, and sexual fantasies. Kinsey could not obtain public funding for the research, so he funded it privately through the Kinsey Institute.

- **1949**

The Soviet Union tests an atomic bomb, marking the beginning of the Cold War.

- **1950s-1960s**

Believing that the Soviet Union had discovered a form of mind control, the CIA starts a covert program called MKUltra with the purpose of developing mind control techniques that could be used for interrogation or brainwashing. MKUltra researchers subjected unwitting participants to psychological torture involving the administration of electric shocks and the psychoactive drug LSD. The program also involved hiring Nazi and Japanese doctors who had performed unethical experiments on living human subjects. The MKUltra program came to the public's attention during Congressional Hearings held from 1975 to 1977.

- **1953**

James Watson (1928-) and Francis Crick (1916-2004) propose a model for the structure of DNA, for which they eventually would share the Nobel Prize with Maurice Wilkins (1916-2004) in 1962. An X-ray diffraction photo of DNA (known as Photo 51) generated by Rosalind Franklin (1920-1958) was crucial for verifying Watson and Crick's model. Wilkins showed Watson and Crick Photo 51 without Franklin's permission. Watson/Crick and Wilkins/Franklin published their papers in the same issue of the journal *Nature* in 1953. Neither Wilkins nor Franklin were named as authors on the Watson/Crick paper (and vice versa). Franklin was not awarded the Nobel Prize because she died in 1958 from ovarian cancer at age 37, and the prize is not awarded posthumously.

- **1954**

The U.S. Atomic Energy Commission revokes Oppenheimer's security clearance based on its determination that, due to his associations with members of the Communist Party, he could not be trusted to be loyal to the U.S. and was a security risk.

- **1956-1980**

Saul Krugman, Joan Giles and other researchers conduct hepatitis experiments on mentally disabled children at The Willowbrook State School. They intentionally infected subjects with a mild form of hepatitis for the purpose of developing a vaccine to a stronger form of the disease, which was endemic at Willowbrook. Children were infected both by injecting them with the hepatitis and making them drink chocolate milk mixed with the feces of people infected with the disease. The experiments were approved by the New York Department of Health. The Willowbrook State School itself was a center of controversy for abuse and neglect of children. In a speech to Congress in 1965, Robert F. Kennedy called the school a “snake pit.”

- **1957**

The Soviets launch Sputnik, the first satellite to orbit the Earth, which triggers the U.S. government to increase its investments in science and technology to avoid falling behind in the space race.

- **1957-1962**

In 1957, thalidomide is marketed in West Germany as medication to treat morning sickness during pregnancy. About 10,000 infants, mostly in West Germany, are born with severe birth defects as a result of exposure to this drug. 2,000 children die from thalidomide exposure. In 1960, Frances Kathleen Oldham Kelsey (1914-2015), a drug reviewer for the FDA, refused to approve the drug. Soon, countries around the world ban the drug. Kelsey is awarded the President's Award for Distinguished Federal Civilian Service in 1962.

- **1961**

President John F. Kennedy commits the U.S. to the goal of putting a man on the moon by the end of the decade.

- **1961**

Rachel Carson (1907-1964) publishes *Silent Spring*, which alerts people to the harmful environmental and public health effects pesticides, especially DDT. Her book launches the environmentalist movement.

- **1961-1962**

Stanley Milgram (1933-1984) conducts his "electric shock" experiments, which proved that people are willing to do things that they consider to be morally wrong when following the orders of an authority. The experiments, which had several variations, included a learner, a teacher, and a researcher. The learner was connected to electrodes. If the learner gave an incorrect response to a question, the researcher would instruct the teacher to push a button on a machine to give the learner an electric shock. Teachers were willing to do this even when the dial on the machine was turned up to "dangerous" levels and the learner were crying out in pain and asking for the experiments to stop. In reality, no shocks were given. The purpose of the experiments was to test subjects' willingness to obey an authority figure. Since then, other researchers who have repeated these experiments have obtained similar results.

- **1963**

Due to fear of the long-lasting impacts of nuclear fallout, dozens of countries signed the Partial Nuclear Test Ban Treaty, which bans nuclear weapons tests, except those conducted underground.

- **1963**

The Public Health Service publishes its *Guide to the Care and Use of Laboratory Animals*, which describes standards, practices, and procedures for conducting experiments with animals and protecting their welfare. The Guide has been revised eight times, most recently in 2011.

- **1964**

The World Medical Association publishes Declaration at Helsinki, Ethical Principles for Research Involving Human Subjects. The Helsinki Declaration has been revised numerous times, most recently in 2013.

- **1964**

The U.S. Surgeon General's office issues its first of several reports on health problems related to smoking.

- **1965**

The Association for Assessment and Accreditation of Laboratory Animal Care (AALAC) is established as an organization that accredits institutions which perform

experiments on laboratory animals. AAALAC evaluates organizations based on their compliance with standards, practices, and procedures described in the Public Health Service's *Guide to the Care and Use of Laboratory Animals*.

- **1966**

Henry Beecher (1904-1976) publishes an article in the *New England Journal of Medicine* describing 22 unethical studies reported in the medical literature. Beecher selected these studies from 186 he had found in medical journals and said they represented a pattern of unethical behavior. Though Beecher did not include the names of researchers and institutions in his article, it is likely that Example 16 was the Willowbrook Experiment and Examples 1-3 were examples of methodologies that had been used in the Tuskegee Study.

- **1966**

The U.S. Congress adopts the Animal Welfare Act, which protect animals used in research, excluding rodents and birds. The Act was adopted partly due to the fear of dogs being stolen to be used in research. During the 1960s, various states adopt or revise animal cruelty laws that protect agricultural, domestic, wild, and laboratory animals.

- **1968**

The United States, Soviet Union, and United Kingdom sign the Treaty on the Nonproliferation of Nuclear Weapons and agree to pursue nuclear disarmament policies. Today, 190 countries have signed the treaty. Notably, India, Pakistan, Israel, and North Korea are not parties to the treaty.

- **1969**

The U.S. lands the first man on the moon, Neil Armstrong.

- **1972**

The U.S., Soviet Union, and dozens of other countries sign the Biological Weapons and Toxins Convention, which prohibits the development, production, acquisition, transfer, stockpiling and use of biological and toxin weapons but allows research on defensive countermeasures to biological weapons and toxins. However, the Soviet Union continues to conduct secret research on offensive bioweapons during the 1970s and 1980s.

- **1973**

After conducting hearings on unethical research involving human subjects, including the Tuskegee study, Congress passes the National Research Act in 1973, which President Nixon signs in 1974. The Act authorizes federal agencies (e.g., the NIH and FDA) to develop human research regulations. The regulations require institutions to form Institutional Review Boards (IRBs) to review and oversee research with human subjects.

- **1973-1975**

Leaders of the emerging field of recombinant DNA research meet in Asilomar, CA to discuss biosafety issues and develop biosafety protocols. In 1974, Paul Berg and other top scientists call for a voluntary moratorium on recombinant DNA experiments until these risks are better understood and appropriate safety protocols are in place. In 1975, Berg and others published a paper in describing some biosafety principles and recommendations for recombinant DNA experiments.

- **1974**

The NIH forms the recombinant DNA advisory committee (RAC) to provide guidance for NIH-funded recombinant DNA experiments. In 1976, the RAC publishes guidelines for recombinant DNA research, which include the requirement that funded institutions establish institutional biosafety committees (IBCs) to oversee recombinant DNA experiments. Although not required by NIH policy, most IBCs also oversee other types of dangerous biological research, such as research involving the collection and storage of deadly pathogens.

- **1974**

William Summerlin admits to fabricating data by using a marker to make black spots on white mice at Sloan Kettering Cancer Institute. He was developing a technique for transplanting skin grafts.

- **1974**

Monsanto Corporation and Harvard University reach a deal for the first major corporate investment in a university.

- **1975**

Peter Singer publishes *Animal Liberation*, which provides a philosophical defense of the animal rights movement. Singer argues that all sentient beings have inherent moral value and that to think otherwise is a form of bias which he calls speciesism, i.e., the view that human beings are inherently superior to other forms of life. Singer argues that many socially accepted ways of using animals, such as for food, sport, or experimentation, are unethical.

- **1975**

Harvard entomologist E.O. Wilson (1929-2021) publishes *Sociobiology*, which reignites the centuries-old "nature vs. nurture" debate. His book proposes biological and evolutionary explanations of human behavior and culture.

- **1978**

The Animal Liberation Front, a radical animal rights group, is formed. This group has engaged in tactics involving raiding labs, destroying lab equipment, releasing animals into the wild, and threatening scientists. The ALF and related groups have caused millions of dollars in property damage.

- **1978**

Louise Brown, the world's first baby conceived by *in vitro* fertilization, is born in the U.K., alive and healthy.

- **1979**

The National Commission for the Protection of Human Subjects in Biomedical and Behavioral Research publishes *The Belmont Report: Principles of Ethical Research on Human Subjects*. The Report articulates three ethical principles for research with human subjects, respects for persons, beneficence, and justice, and provides a conceptual foundation for a major revision of the U.S. federal research regulations in 1981.

- **1980**

Congress passes the Bayh-Dole Act, which allows researchers to patent inventions developed with government funds; the Act is amended by the Technology Transfer Act in 1986.

- **1980**

In *Diamond v. Chakrabarty*, the U.S. Supreme Court rules that a genetically modified bacterium can be patented because it is the product of human ingenuity. This sets a precedent for patents on other life forms and helps to establish solid intellectual property protection for the new biotechnology industry.

- **1981**

The Whitehead Institute is established at MIT, which represents a major private investment in a university. Other universities follow this example and begin forming complex partnerships with industry.

- **1981**

The Department of Health, Education, and Welfare (a precursor to the Department of Health and Human Services, DHHS) conducts major revisions of the federal human research regulations for human subjects research.

- **1981**

John Darsee, a postdoctoral fellow at Harvard, is accused of fabricating data. 17 of his papers were retracted.

- **1982**

William Broad and Nicholas Wade publish *Betrayers of Truth*. The book claims that there is more misconduct in science than researchers want to admit and suggests that famous scientists, including Isaac Newton, Gregor Mendel, and Robert Millikan were not completely honest with their data. Their book helps to launch an era of "fraud busting" in science.

- **1984**

Members of a religious cult led by Bhagwan Rajneesh sprayed salmonella bacteria on salad bars in ten restaurants in The Dalles, Oregon in an effort to make people too sick to vote in Wasco County elections so that the cult's candidates would win. 751 people developed salmonella poisoning, but fortunately no one died. This is the first known bioterrorist attack on U.S. soil.

- **1984-1993**

After collaborating with Robert Gallo on isolating HIV from biological samples provided by AIDS patients, Luc Montagnier accuses Gallo of misappropriating an HIV strain and accuses him of research misconduct, for which Gallo is found innocent. Gallo and Montagnier also have a dispute about who should be credited with discovering HIV and who can patent a test for the virus. The U.S. and French governments reach an agreement to settle the controversy.

- **1985**

The Animal Welfare Act is amended to create the Animal Welfare Information Center, the purpose of which is to improve the public's access to animal welfare information and to develop more humane methods of animal research. The amendment also requires that government funded animal research be reviewed by Animal Care and Use Committees.

- **1986**

Aerospace engineer Roger Boisjoly warns NASA about possible O-ring failure during the Space Shuttle Challenger launch, due to cold weather. The O-rings, which are made of rubber, are designed to function properly at temperatures as low as 32° F (0° C), but the predicted air temperature at launch time was 26° F (-3° C). Boisjoly meets with NASA officials to discuss the problem, but they decide to go ahead with the launch, and the Challenger explodes, killing all seven crew members. A special committee investigating the accident determines that the O-rings did not seal properly due to the cold weather, which allowed rocket fuels to leak and explode. Nobel Prize-winning physicist Richard Feynman plays a key role in convincing the committee that the explosion was due to O-ring failure.

- **1987**

A NIMH panel concludes that Steven Breuning fabricated and falsified data in 24 paper. Breuning is convicted of defrauding the federal government in 1988.

- **1987**

Martin Luther King is accused of plagiarizing his Ph.D. dissertation.

- **1987-1996**

Margot O'Toole, a post-doctoral student at the Whitehead Institute, has some questions about data in a paper authored by six of her colleagues and published in

the journal *Cell* in 1986. She asks to examine Thereza-Imanishi-Kari's lab notebooks, which seem to be inconsistent with published results. She accuses Imanishi-Kari of fabricating and falsifying data. The ensuing investigation leads to inquiries by MIT and Tufts as well as the NIH and a Congressional committee chaired by Rep. John Dingell. Nobel Prize winner David Baltimore is one of the co-authors on the disputed paper. Although he was not accused of misconduct, Baltimore resigns as President of Rockefeller University. He described the investigation, which was covered by the New York Times, as a "witch hunt." An appeals board at the DHHS eventually exonerated Imanishi-Kari, who admitted only to poor record keeping.

- **1988**

Harvard and Dow Chemical patent a genetically engineered mouse used to study cancer.

- **1989**

The PHS forms two agencies, the Office of Scientific Integrity and the Office of Scientific Integrity Review to investigate scientific misconduct and provide information and support for universities. It also amends its definition of misconduct. The two agencies are reorganized in 1992 as the Office of Research Integrity (ORI).

- **1989**

The NIH requires that all graduate students on training grants receive education in responsible conduct of research. In the ensuing years, these requirements are expanded to include post-doctoral trainees and intramural researchers. NSF adopted RCR training requirements in 2010.

- **1989**

Stanley Pons and Martin Fleischmann hold a press conference at the University of Utah to announce that they have discovered a way to produce nuclear fusion at room temperatures. After dozens of labs across the world fail to reproduce their results, they are accused of fraud, sloppiness, and self-deception.

- **1989**

The National Academy of Science (NAS) publishes *On Being A Scientist* (revised in 1994 and 2009), which is a free, short book on research ethics for scientists in training.

- **1990**

The U.S. launches the Human Genome Project, a \$20 billion effort to map and sequence the human genome.

- **1990**

W. French Anderson begins the first human gene therapy clinical trial on patients with ADA deficiency, a genetic disease that affects the immune system.

- **1990**

In *Moore v. Regents of the University of California*, the California Supreme Court rules that researchers have intellectual property rights in a cell line derived from Moore's tissue, but that Moore did not have any property rights in his own tissue. The Court also rules that the researchers violated Moore's right to informed consent by not disclosing their commercial interests in his tissue sample to him. Most courts have followed this ruling by holding that patients relinquish their property rights to tissues when they donate them to research or when they are leftover ("abandoned") during surgery of medical procedures.

- **1990**

Congress investigates conflicts of interest (COIs) involving Pharmatec and the University of Florida and other COIs in biomedical research.

- **1990s-present**

Europeans oppose the introduction of genetically manipulated foods and crops. Consumers in the U.S. are more receptive to GM plants and animals. After banning GM crops in 1998, the European Union allows the cultivation of GM crops but requires GM foods to be labeled as such. In 2022, the U.S. mandates the labelling of GM foods.

- **1991**

U.S. federal agencies revise their human research regulations. All U.S. government agencies now accept one basic regulatory framework, known as "the Common Rule" (45 CFR 46).

- **1992**

NAS publishes *Responsible Science: Ensuring the Integrity of the Research Process*. The book estimates the incidence of misconduct, discusses some of the causes of misconduct, proposes a definition of research misconduct, and recommends some strategies for preventing misconduct and promoting research integrity.

- **1992**

The Public Health Service (PHS), which funds NIH research, consolidates the Office of Scientific Integrity and the Office of Scientific Integrity Review into the Office of Research Integrity (ORI). ORI develops policies, procedures, policies, and regulations for preventing, reporting, and investigating research misconduct; reviews and monitors misconduct investigations conducted by PHS-funded institutions and makes recommendations concerning misconduct findings and administrative; provides technical assistance to institutions that are responding to misconduct allegations; and supports research, conferences, and education on the responsible conduct of research.

- **1993**

In *Daubert v. Merrell Dow Pharmaceuticals* the U.S. Supreme Court rules that federal judges serve as the gatekeepers for admitting scientific testimony in court and that they can use a variety of criteria, including testability, reliability, peer review, and general acceptance for determining whether testimony is scientific. The intent of the ruling was to allow cutting-edge science that might not be generally accepted to be admitted as evidence in the courtroom. Prior to *Daubert*, federal judges followed the general acceptance standard articulated in *Frye vs. United States* (1923). State courts follow either the *Daubert* or *Frye* standard, depending on the jurisdiction.

- **1993**

Fertility researchers successfully clone human embryos.

- **1994**

Harvard psychologist Richard Herrnstein and Charles Murray publish *The Bell Curve*, a controversial book that reignites the centuries old debate about biology, race, and intelligence.

- **1994**

Roger Poisson admits to fabricating and falsifying patient data in NIH-funded breast cancer clinical trials in order allow his patients to qualify for enrollment and have access to experimental treatments.

- **1994**

The NIH applies for patents on thousands of gene fragments in order to undercut private efforts to patent gene fragments. The Patent Office rejects the NIH's applications because they did not clearly define a practical use for the fragments.

- **1994-1995**

A Commission chair by Kenneth Ryan, convened by NIH, holds meetings on defining, investigating, and preventing scientific misconduct.

- **1994**

The Clinton Administration declassifies information about secret human radiation experiments conducted from the 1940s-1980s and issues an apology.

- **1994**

Two scientists who worked at Philip Morris, Victor DeNobel and Paul Mele, testify before Congress about secret research on the addictive properties of nicotine. If the research had been made public, the FDA or Congress might have taken additional steps to regulate tobacco as a drug. Many states and individuals brought litigation against tobacco companies, which led to a \$206 billion settlement between tobacco companies and 46 states. The scientific community also publishes more data on the dangers of second-hand smoke.

- **1995**

Boots Pharmaceuticals pressures Betty Dong to withdraw a paper from publication in *JAMA* showing that its drug, Synthroid, is not more effective than generic equivalents at treating hypothyroidism.

- **1995-2005**

Dozens of studies are published in biomedical journals which provide data on the relationship between the source of research funding and financial interests and the

outcomes of research studies in the biomedical sciences, and the close relationship between academic researchers and the pharmaceutical and biotechnology industries.

- **1995**
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The NIH and NSF revise their conflict of interest policies.

- **1995**
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Scientists and defense analysts become concerned about the use of chemical or biological weapons by a terrorist group after Aum Shinrikyo, a Japanese doomsday cult, releases sarin gas in a Tokyo subway, killing 12 people and sending 5,500 to hospitals. The group also attempted (unsuccessfully) to spray anthrax spores over Tokyo. In 1998, terrorism experts warn about the use of biological or chemical weapons by Osama bin Laden and Saddam Hussein.

- **1995**
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Over 200 religious leaders, led by biotechnology critic Jeremy Rifkin, protest the patenting of plants, animals, and human body parts in Washington, D.C.

- **1996**
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Dolly, the world's first cloned sheep, is born; her birth is announced in 1997. Several European nations ban human cloning. Congress considers a bill to ban all human cloning but decides not to after scientists argue that the bill would undermine biomedical research.

- **1996**
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Physics professor Alan Sokal submits a paper titled "Transgressing the Boundaries: Towards a Transformative Hermeneutics of Quantum Gravity" to *Social Text*, a cultural studies journal. Sokal filled the paper with errors of fact and reasoning and outright nonsense that any well-qualified physicist would be able to spot in order to test the rigor of the journal's peer review process. The paper proposed that quantum gravity does not exist independently of human beings and is a social and linguistic construct. After the paper was accepted for publication, Sokal revealed the hoax.

- **1997**

The ICMJE, representing over 400 biomedical journals, revises its authorship guidelines. The ICMJE recommends that to be an author on a scientific paper one must make a substantial contribution to research design, data collection, data analysis, or data interpretation and write the paper or critically revise and review it.

- **1997**

In an article published in *New England Journal of Medicine*, Peter Lurie and Sidney Wolfe accuse the NIH, WHO, UN and CDC of designing and conducting unethical clinical on the prevention of mother-child transmission of HIV in developing countries, because the trials include placebo control groups even though the experimental treatment had been tested in western nations and proven effective. Representatives NIH Director Harold Varmus and CDC Director David Satcher argue that placebo control groups were needed to ensure scientific rigor because the dose of the experimental treatment being tested in the trials was much lower than the dose that had been proven effective. The dispute spurs a closer examination of international research ethics codes and guidelines.

- **1998**

Scientists perfect methods for growing human embryonic stem cells. Some countries ban the research; others promote it.

- **1998**

Craig Venter forms Celera Genomics and begins a private effort to sequence the human genome, using dozens of automated sequencing machines.

- **1998-1999**

Apotex forces Nancy Olivieri, a clinical researcher at the University of Toronto, to withdraw a paper that exposes safety concerns about its drug deferiprone, which is used to treat thalassemia. The company tries to discredit Olivieri and have her fired.

- **1999**

Eighteen-year-old Jessie Gelsinger dies in a human gene therapy experiment at the University of Pennsylvania. The event triggers heightened scrutiny of conflicts of interest in human subjects research, including institutional conflicts of interest. Penn settles a lawsuit brought by the Gelsinger family for an undisclosed amount of money.

- **1999-2009**

Human research lawsuits increase dramatically. Alan Milstein, from the law firm Sherman, Silverstein, Kohl, Rose & Podolsky, P.A., instigates 13 lawsuits against researchers, universities, pharmaceutical companies, and Institutional Review Board members.

- **1999**

The U.S. NIH and OHRP require all people conducting or overseeing human subjects research to have training in research ethics.

- **2000**

The U.S. Office of Science and Technology Policy finalizes a federal definition of misconduct as "fabrication, falsification or plagiarism" but not "honest error or differences of opinion." Misconduct must be committed knowingly, intentionally, or recklessly.

- **2000**

ORI proposes mandatory training in responsible conduct of research (RCR) for all researchers on PHS grants, including junior senior investigators, students, and technicians. Several scientific associations and universities oppose the policy as an unnecessary and un-funded mandate. The Bush Administration suspends the ORI proposal in 2001 on the grounds that the agency failed to follow proper procedures for proposing new government regulations. Many research institutions voluntarily expand their RCR training programs.

- **2001**

Celera and the Human Genome Project both finish 99% complete drafts of the human genome and publish their results in *Science* and *Nature*.

- **2001**

Congress debates legislation on human cloning but does not adopt any laws.

- **2001**

Several journals, including *Nature* and *JAMA*, experiment with requiring authors to describe their responsibilities (e.g., designed experiments, collected data, analyzed data, wrote a first draft of the paper, etc.) when publishing research. Today, many journals follow this policy.

- **2001**

The Bush Administration announces that the NIH will only fund human embryonic stem cell research on approximately 64 cell lines created from leftover human embryos.

- **2001**

In the fall of 2001, Bruce Ivins, a biodefense researcher at the U.S. Army Medical Research Institute of Infectious Diseases in Fort Detrick, allegedly sent letters laced with anthrax spores to Senators Tom Daschle and Patrick Leahy and several members of the media. The attacks, which killed five people, sickened 17, and exposed dozens of postal workers to anthrax spores, created tremendous anxiety in a nation already reeling from the terrorist attacks of September 11, 2001. The Federal Bureau of Investigation's lengthy and costly investigation eventually named Ivins as the suspect in 2008, but Ivins committed suicide before he could be apprehended and brought to trial.

- **2002**

Bell Labs determines that Jan Hendrick Schön, a rising star working in condensed matter physics and nanotechnology who published dozens of articles in a short period of time in prestigious journals, had fabricated and falsified data. 28 papers authored by Schön were retracted.

- **2002**

The President's Council on Bioethics recommends that the U.S. ban reproductive cloning and enact a moratorium on research cloning.

- **2002**

Historian Stephen Ambrose is accused of plagiarism.

- **2002**

The NAS publishes *Integrity in Scientific Research*, which recommends that universities develop programs for education in responsible conduct of research (RCR) as well as policies and procedures to deal with research ethics.

- **2002**

North Korea declares that it has a secret nuclear weapons program and warns that it has other "more powerful" weapons.

- **2002-2004**

Scientists publish several papers in prominent journals with direct implications for bioterrorism. A paper published in the *Journal of Virology* described a method for genetically engineering a form of mousepox virus that is much deadlier than the naturally occurring strain. A paper published in *Science* showed how to make the poliovirus by obtaining supplies from a mail-order company. A paper published in *PNAS* develop a mathematical model for showing how many people would be killed by infecting the U.S. milk supply with botulinum toxin. In 2003, the American Society for Microbiology (ASM), the National Academy of Sciences, and the Center for Strategic and International Studies held a meeting to discuss the censorship biological research that poses security risks. Journals agree to self-censor some research.

- **2003**

The U.S. invades Iraq with the stated purpose of eliminating its chemical, biological, and nuclear weapons programs. The U.S. found evidence of weapons programs but no actual weapons.

- **2004**

The EPA suspends the CHEERS study due to criticism from advocacy groups and members of Congress, who claimed that the study was intentionally exposing children to pesticides and targeting minority groups. The EPA revised its human subjects rules in response to a Congressional mandate to strengthen protections for children and pregnant or nursing women.

- **2004**

Ronald Reagan, Jr. makes a presentation in support of federal funding for embryonic stem cell research to the Democratic Convention. Stem cell research (and therapeutic cloning) become hot issues in the 2004 Presidential election.

- **2004**

Merck withdraws its drug Vioxx from the market, due to safety and liability issues. As many as 50,000 people had a heart attack or stroke while take the drug, and thousands sued the company. As early as 2001, Merck scientists suspected that Vioxx could increase the cardiovascular risks, but researchers funded by Merck did not publish some of the data that would support these suspicions, even though they reported it to the FDA. In 2001, the FDA warned Merck that it had misrepresented Vioxx's safety profile to the public and in 2002 it issued a black box warning for the drug. A systematic review of antidepressant medications known as selective serotonin uptake inhibitors (SSRIs) found that some of these drugs increase the risks of suicide in adolescents and children. The review included data from the U.K.'s Committee on Safety in Medicines, which had not been previously published. Patients, parents, researchers, and policymakers accused companies intentionally hiding this data from the public, and New York Attorney General Eliot Spitzer sued Glaxo for fraud. As a result of these problems related to data suppression, government agencies (including the FDA) and journals now require clinical trials to be registered on a publicly available website. Registration includes important information about the studies, including research design, interventions, and methods; research sites and personnel; contact information; and research results (but not raw data).

- **2005**

In response to criticism from Congress, the NIH revises its conflict of interest rules for employees. The rules place restrictions on ownership of stock in substantially affected organizations (such as pharmaceutical or biotech companies) by employees and prohibit employees with consulting with industry for pay.

- **2005**

Seoul University research Woo Suk Hwang admits to fabricating data in two papers published in the journal *Science*. In the papers, Hwang claimed that he had used nuclear transfer techniques to develop patient-specific human embryonic stem cells.

- **2005**

University of Vermont researcher Eric Poehlman admits to fabricating or falsifying data in 15 federal grants and 17 publications. Poehlman served a year and day in federal prison and agreed to pay the U.S. government \$180,000 in fines.

- **2005**

In response to recommendations from a National Research Council report titled “Biotechnology in the Age of Terrorism,” the Department of Health and Human Services establishes the National Science Advisory Board for Biosecurity (NSABB) to provide advice and guidance to federal agencies, scientists, and journals concerning oversight and public of research in biotechnology or biomedicine which can be readily applied to cause significant harm to public health, agriculture, the economy, or national security (i.e. “dual use” research).

The Obama Administration announces it will significantly expand NIH funding of human embryonic stem cell research which had been restricted under the Bush Administration.

- **2010**

The National Science Foundation (NSF) announces RCR training requirements for funded investigators, students, and trainees. The NIH expands and strengthens its RCR training requirements.

- **2010**

While doing research on the Tuskegee Syphilis Study, Susan Reverby, Professor of Women’s Studies at Wellesley College, uncovered documents concerning unethical research experiments on human subjects conducted by the U.S. government in Guatemala from 1946 to 1948. The research involved intentionally infecting over 1300 subjects with syphilis to test the effectiveness of penicillin in preventing this disease. Only 700 subjects were given penicillin and 83 died as a result of the study. The subjects were not informed that they were participating in an experiment.

- **2010**

Lancet retracts a paper, published in 1998 by Andrew Wakefield and colleagues, linking the measles, mumps, and rubella vaccine to autism. *Lancet* retracted the paper after an investigation by journalist Brian Deer found that Wakefield had not disclosed a significant financial interest and had not obtained ethics board approval for the study. Wakefield’s research had been supported by a law firm that was suing vaccine manufacturers, and lawyer for the firm had helped Wakefield recruit

patients. Wakefield did not disclose his relationship to the law firm in the 1998 paper. In 2010, the U.K.'s General Medical Council revoked Wakefield's license to practice medicine. In 2011, Deer published an article in the British Medical Journal accusing Wakefield of fabricating and falsifying data in the 1998 paper. Deer based his findings on discrepancies between data reported in the paper and the data from patient records.

- **2010**

Jeffrey Beale publishes a list of what he calls "predatory journals." Predatory journals are profit-driven journals that charge high fees for open access publication, promise rapid publication, and have poor (or nonexistent) standards for peer review. Beale later withdraws his list due to legal pressure from journals.

- **2010**

Ivan Oransky and Adam Marcus launch Retraction Watch, a blog that post retractions of scientific papers and articles related to research integrity.

- **2010**

The World Conference on Research Integrity releases the Singapore Statement on Research Integrity, a code of ethics for scientists in various disciplines.

- **2011**

The NIH and NSF revise their conflict of interest rules for funded research.

- **2011**

Journalist Rebecca Skloot publishes a widely-acclaimed book about Henrietta Lacks, an African American woman who provided the tissue for a widely-used cell line known as HeLa (an abbreviation of her name). In 1951, Lacks underwent treatment for cervical cancer at Johns Hopkins Hospital and died later that year. Researchers discovered that they were able to culture the cells from Lacks' tumor and keep them alive, which was the first time that scientists had been able to grow a human cell line. HeLa cells have been used in thousands of laboratories around the world in biomedical experiments. Skloot was interested in finding out where the HeLa cell line came from, and she discovered that it came from Henrietta Lacks. Skloot interviewed Lacks' family and learned that researchers had grown her tumor cells without her consent and without providing the family any compensation, which was a common practice at that time. Skloot decided to donate profits from her book to a

private foundation she started whose purpose is to raise awareness about the role the human biological materials play in research and issues related to consent and ownership. In 2013, the NIH reached an agreement with Lacks' family concerning access to genomic data from NIH-owned HeLa cell lines. The agreement gives the family control over access to the data and acknowledgment in scientific papers. In 2021, the Lacks family sued Thermo Fisher Scientific, a company that commercialized the the cell line. The lawsuit claims that the company unjustly profited from Lacks' tissue without her consent.

- **2011-2013**

Several authors publish papers documenting a dramatic increase in the number of retracted papers since 2001 and that the majority of the retractions are due to research misconduct.

- **2012**

Two papers embroiled in controversy were published in *Science* and *Nature* after months of debate about their implications for bioterrorism. The papers reported the results of NIH-sponsored research conducted by a team working in the Netherlands, led by Ron Fouchier, and a team working at the University of Wisconsin, led by Yoshihiro Kawaoka. The researchers were able to genetically modify an H5N1 avian flu virus so that it can be transmitted by air between mammals. Currently, avian flu can only be contracted through direct contact with birds. The virus is highly lethal, with a case fatality rate of over 50%. Over 300 people have died from the virus since 1997. The National Science Advisory Board for Biosecurity (NSABB) initially recommend that the papers be published in redacted form, with key details removed and only made available to responsible scientists, so the terrorists or others could not use the information to make deadly bioweapons. However, the NSABB changed its mind and recommended full publication of both papers after learning more about the value of the research for public health (e.g., monitoring of bird populations, vaccine development), biosafety measures for containing the virus, how difficult it would be for terrorists to replicate the work, and legal problems with redacted publication.

- **2013**

The NIH launches the reproducibility initiative in response to published reports of problems with the reproducibility of biomedical and social/behavioral research. The initiative includes guidelines and educational materials for promoting rigor, transparency, openness, and reproducibility in research and reducing bias and error. Scientific journals also develop policies designed to promote reproducibility.

- **2013**

In Association for *Molecular Pathology et al. v. Myriad Genetics*, the U.S. Supreme Court rules that isolated and purified DNA cannot be patented. Only DNA that has been modified by human beings can be patented. The ruling invalidates Myriad's patents on BRCA1 and BRCA2 genes and creates uncertainty concerning the legal validity of other types of patents on isolated and purified chemicals, such as chemicals derived from medicinal plants.

- **2014**

Haruko Obokata, a biochemist at the RIKEN Center for Developmental Biology in Kobe, Japan, and coauthors published two high-profile papers in *Nature* describing a method for converting adult spleen cells in mice into pluripotent stem cells by means of chemical stimulation and physical stress. Several weeks after the papers were published, researchers at the RIKEN Center were unable to reproduce the results and they accused Obokata, who was the lead author on the papers, of misconduct. The journal retracted both papers in July after an investigation by the RIKEN center found that Obokata had fabricated and falsified data. Later that year, Obokata's advisor, Yoshiki Sasai, committed suicide by hanging himself.

- **2014**

The Department of Health and Human Services (DHHS) announces a pause on new funding of gain of function (GOF) genetic manipulation experiments involving SARS, MERS, and other potential pandemic pathogens (PPPs) and asks the NSABB to review the issues and make policy recommendations. GOF research is research that attempts to genetically manipulate a pathogen to enable it to acquire a new function, such as increased transmissibility or virulence. The NSABB recommended that the DHHS should lift the funding pause after implementing an oversight framework that minimizes and manages the risks of GOF research with PPPs, which DHHS did in 2017.

- **2015**

A research team led by Ralph Baric at the University of North Carolina publishes a paper in *Nature Medicine* reporting the results of an experiment in which they used a mouse-adapted SARS-CoV backbone to create a chimeric virus that expresses the spike protein of SHC014. The virus was able to use angiotensin-converting enzyme 2 (ACE2) receptors to infect and replicate in human airway cells and caused pathogenesis in mouse cells. The authors hypothesized that coronavirus pools found in horseshoe bats maintain spike proteins that give them the capability of infecting humans and stated that there is "a potential risk of SARS-CoV re-emergence from

viruses currently circulating in bat populations.” Baric’s research was funded by the NIH. Shi Zengli, the second to last author on the paper, provided the genomic sequence for the spike protein. Shi conducted similar experiments at the Wuhan Institute of Virology. Baric’s lab was Biosafety Level 3, but Shi’s was only Biosafety Level 2. The paper became embroiled in controversy in 2021 when U.S. Senator Rand Paul alleged that it violated DHHS’s funding pause on GOF research.

- **2016**

The NIH places a temporary moratorium on funding for experiments involving human-animal chimeras while it revises existing policies that govern this research. The NIH lifted the moratorium the following year.

- **2017**

17 federal agencies publish the Final Rule for revisions to the Common Rule, which becomes effective in 2019. The revisions clarify and expand categories are research that are exempted from the Common Rule, enhance informed consent requirements, relax the need for continuing review once a study has stop recruiting participants, and requires a single IRB be responsible for reviewing multisite research.

- **2017-2019**

Drawing inspiration from the Sokal hoax of 1996, in 2017-2018, philosopher Peter Boghossian, mathematician James Lindsay, and social commentator Helen Pluckrose submit bogus papers to journals to test their peer review standards. Several of the papers were published. The authors revealed their hoax in 2019. Boghossian and Pluckrose authored a paper, which was published in *Cogent Social Sciences*.

- **2018**

In October, He Jiankui, a scientist of the Southern University of Science and Technology in Shenzhen, China, announces the birth of the world’s first gene edited babies, both girls. He claims that he used CRISPR-Cas 9 technology to modify the CCR5 gene to give the girls immunity to HIV. The announcement generates outrage around the world and many scientists and policymakers call for a ban on creating gene edited babies.

- **2020**

During the COVID-19 pandemic, which killed millions of people around the world, scientists worked feverishly to conduct research that could help combat this public

health crisis. To make data and results available as quickly as possible, many journals expedited their peer review process for COVID-19 submissions and scientists published articles on pre-print servers before undergoing peer review. Unfortunately, the rush to publish led to the dissemination of invalid, irreproducible, and falsified results in some cases. Dozens of COVID-19 papers had to be retracted or withdrawn. Two prominent biomedical journals, the *New England Journal of Medicine* and the *Lancet*, retracted papers after the editors learned that the healthcare analytics company Surgisphere, which provided the data used in the studies, was not making the raw data available to independent scientists who found inconsistencies in the data and wanted to do an audit. The journals also learned that Surgisphere did not make all the data available to the scientists who authored the papers.

- **2022**

In early 2020, at the beginning of the COVID-19 pandemic, US scientists claimed that the SARS-CoV-2 virus that causes the disease probably moved from horseshoe bats into the human population via an undetermined intermediate host species. Others believed that SARS-CoV-2 was a genetically engineered pathogen that had escaped from a laboratory at the Wuhan Institute of Virology (WIV) in China. The World Health Organization attempted to investigate the origins of the virus by interviewing scientists at WIV and inspecting facilities, but they were unable to obtain all the information they needed. During 2020, there was a scientific consensus that SARS-CoV-2 probably had a natural origin. Opinions began to change in 2021, however, when highly respected science writer Nicholas Wade published several articles critically examining the two competing hypotheses: natural origin vs. laboratory escape. Wade argued that there was substantial evidence supporting the laboratory escape hypothesis, because an intermediate host species had not been found despite intense searching and a key part of the protein that the virus uses to infect cells, known as the furin cleavage site, does not occur in related coronaviruses. After Wade's articles were published, many scientists started publishing papers examining evidence concerning the origins of SARS-CoV-2. However, the origins of SARS-CoV-2 remain a mystery.

- **2023**

Neuroscientist Marc Tessier-Lavigne resigns his position as President of Stanford University after a special committee appointed by the Board of Trustees found that he failed to retract or correct 12 papers with serious flaws in a timely fashion. Tessier-Lavigne was a principal author on five of the papers, four of which included fabricated or falsified data or images. The Committee did not find that Tessier-Lavigne engaged in research misconduct. Commentators on the PubPeer website, which provides post-peer review of scientific papers, had alleged in November 2022 that digital images in the papers had been duplicated or otherwise manipulated.

